

Julien Ferry

Postdoctoral Researcher at Polytechnique Montréal

PhD in Computer Science and Telecommunications | Engineer's Degree, INSA Toulouse – Computer Science and Networks

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Education

October 2020 - October 2023	LAAS-CNRS (Toulouse, France) <i>PhD (Université Toulouse III Paul Sabatier) - Defended on the 9th of October 2023</i> <u>Title:</u> Addressing interpretability, fairness and privacy in machine learning through combinatorial optimization methods <u>Advisors:</u> Marie-José Huguet (LAAS-CNRS), Sébastien Gambs (Université du Québec à Montréal) and Mohamed Siala (LAAS-CNRS).
2015 - 2020	INSA Toulouse <i>Engineer's degree in Electrical and Computer Engineering – Major: Distributed Systems and Big Data</i> Ranked 1st in graduating class
Autumn 2017	The University of Texas at Austin <i>Study semester abroad</i> University Honors Fall Semester 2017

Professional Experience

Since January 2024	Polytechnique Montréal <i>Postdoctoral researcher - SCALE-AI Chair in Data-Driven Supply Chains (Prof. Thibaut Vidal), Mathematics and Industrial Engineering Department</i> Use of combinatorial optimization methods to address several dimensions of trustworthy machine learning, including: • privacy risks associated with interpretable or structured machine learning models and post-hoc explanations of their predictions • defense mechanisms against these risks • characterization of tradeoffs between accuracy, fairness and sparsity within sets of good models • optimal training or compression of ensemble models • efficient computation of optimal or near-optimal explanations of model decisions • manipulation-proof oblivious algorithmic auditing
November 2023	LAAS-CNRS <i>CNRS research engineer</i> ROC (Operations Research, Combinatorial Optimization and Constraints) team Use of combinatorial optimization for (optimal) machine learning methods
October 2020 - October 2023	LAAS-CNRS <i>PhD student - Defended on the 9th of October 2023</i> Doctoral contract from the University of Toulouse III - Paul Sabatier (funded by the French Ministry for Higher Education and Research) from October 2020 (included) to September 2023 (included), CNRS doctoral contract for October 2023

February - July 2020	Thales Alenia Space (Toulouse) <i>Internship (6 months)</i> • State of the art on bias in machine learning and the existing techniques to detect, evaluate, and mitigate it • Implementation of a generic bias mitigation technique for convolutional neural networks • Conception of a Python module gathering several tools to analyze and correct bias in machine learning • Use of this module on different synthetic and industrial use cases of classification and semantic segmentation
June - September 2019	LAAS-CNRS (Toulouse) <i>Internship (3 months)</i> • State of the art on the different fairness notions and metrics (in machine learning) • Conception and implementation of a supervised learning algorithm producing fair and interpretable models • Visiting student during two weeks at UQAM (Université du Québec à Montréal)
Summers 2015, 2016, 2018	Start People (Rodez) <i>Student job - temp worker</i> • Soulié Restauration (agro-food factory): production agent and machine operator on an automatic casing machine at the end of a training course • Handler (Merico-DeltaPrint, DPD, removals)
June - July 2017	Carrosserie Ruthénoise (Rodez) <i>Student job - secretary</i> Drafting of the Single Document for the Assessment of Occupational Risks and the Chemical Risk Study
July 2016	INSA Toulouse humanitarian center <i>Humanitarian mission - Arequipa (Peru)</i> Construction of a cosmetology workshop in a trade school for disadvantaged young people

Honors

December 2025	Spotlight of our article [KFV25] at the NeurIPS 2025 conference (top 3% articles)
January 2025	Our work on dataset reconstruction attacks against Random Forests [Fer+24b] was nominated as one of Québec Science's Top 10 Scientific Discoveries of 2024
July 2024	Our article [Fer+24b] was selected for oral presentation at the ICML 2024 conference (top 1.5% articles)
November 2021	SIGIR Student Travel Award (CIKM 2021) - ACM Special Interest Group on Information Retrieval
October 2020 - October 2023	Doctoral grant (concours) - French Ministry for Higher Education and Research About 67 000€ (109 300 CAD) for three years

Teaching

April 2025	University of Québec at Montréal (UQAM) • Invited Lecture (master's students) - Privacy and security challenges of AI (3h)
November 2024	INSA Toulouse • Invited Lecture (master's students) - Privacy risks of machine learning models (1h)

2022-2023	INSA Toulouse • Bachelor's Classes – Tutorial Classes - Algorithmic and programming II (11.25h) – Lab Sessions - Algorithmic and programming II (22h) • Master's Classes – Lab Sessions - Supervised and Unsupervised machine learning (11h) – Lab Sessions - Machine learning (8.25h) – Lab Sessions - Integrator project (11h)
2021-2022	INSA Toulouse • Bachelor's Classes – Tutorial Classes - Algorithmic and programming II (11.25h) – Lab Sessions - Algorithmic and programming II (11h) – Lab Sessions - Web programming (8.25h) • Master's Classes – Tutorial Classes - Algorithmic complexity (3.75h) – Lab Sessions - Supervised and Unsupervised machine learning (13.75h) – Lab Sessions - Machine learning (8.25h) – Lab Sessions - Integrator project (11h)
2020-2021	INSA Toulouse • Bachelor's Classes – Tutorial Classes - Algorithmic and programming II (11.25h) – Lab Sessions - Algorithmic and programming II (11h) – Lab Sessions - Web programming (8.25h) – Lab Sessions - Graphs and algorithms (19.25h) • Master's Classes – Tutorial Classes - Algorithmic complexity (7.5h) – Lab Sessions - Integrator project (8.25h)

Languages

- **English:** C1 level (TOEIC score: 955)
- **Spanish:** A2/B1 level (occasional practice)
- **French:** native language

Scientific Skills

- **Machine Learning:** Statistics and probabilities, supervised and unsupervised learning - use of Keras (TensorFlow) and Scikit-learn
- **Big Data:** Conception and use of Big Data platforms (storage/computation)
- **Optimization:** Numerical methods, linear optimization (reals, integers, mixed), constraint programming (Mistral, OR-Tools), metaheuristics
- **Symbolic AI and reasoning:** Logic, combinatorial games, Boolean satisfiability
- **Algorithm design:** algorithmic complexity, data structures, imperative programming, object-oriented programming, functional programming, declarative programming
- **Programming languages:** Python, Java, C, OCaml, Ada, Prolog, Assembly
- **Networks and cybersecurity:** Knowledge and use of the main protocols, socket programming
- **IT infrastructures and architectures:** Hardware architecture (2019: design of an instruction set, a C compiler and VHDL implementation of the associated processor), virtualization, distributed systems
- **Project management:** Use of versioning tools (Git), teamwork, LaTeX
- **Scientific computing:** Use of scientific computing platforms (based on Slurm), parallel computing with MPI and OpenMP, GPU programming and JIT compilation in Python using Numba

- **Trustworthy machine learning:** Implementation of differentially private methods, modeling of fairness constraints, interpretable machine learning algorithms, computation of post-hoc explanations, building or characterization of Rashomon sets

Open-Source Software

The source code required to reproduce the results of each of my research papers is systematically made publicly available, often as user-friendly software libraries. The most notable examples are listed below:

- Contributed to refactoring the codebase and integrating new methods [Kho+26; Tan+26] into **OCEAN**, a library for generating counterfactual explanations for tree ensembles:
 - Source code: <https://github.com/vidalt/OCEAN>
 - Quick setup: `pip install oceanpy`
- **DRAFT** (Dataset Reconstruction Attack From Trained ensembles) Python class [Fer+24b]: implementation of an inference attack targeting reconstruction of a Random Forest's training set
 - Source code: <https://github.com/vidalt/DRAFT>
- **FairCORELS** Python module [Aiv+21]: implementation of a supervised machine learning algorithm producing inherently interpretable models meeting given statistical fairness constraints (supporting common fairness metrics from the literature)
 - Source code: <https://github.com/ferryjul/fairCORELS>
 - Quick setup: `pip install faircorels`
- **FairCORELSv2** Python module [Fer+22b]: enhanced version of the **FairCORELS** module, leveraging an ILP approach and a new data structure to prune the algorithm's search space efficiently and learn interpretable and fair models with optimality guarantees
 - Source code: <https://github.com/ferryjul/fairCORELSv2>
 - Quick setup: `pip install faircorelsv2`
- **HybridCORELS** Python module [FLA24]: implementation of a supervised learning algorithm producing hybrid interpretable models (composed of an interpretable and a black-box part (agnostic) handling the most difficult examples)
 - Source code: <https://github.com/ferryjul/HybridCORELS>
 - Quick setup: `pip install HybridCORELS`
- Integration within two fair supervised learning algorithms of the literature of our **exact and heuristic approaches targeting a better generalization of statistical fairness, leveraging a sample-robust optimization method** [Fer+23c]
 - Source code: <https://github.com/ferryjul/FairnessSampleRobustness>
- **ReconstructionCorrector** Python class [Fer+23b]: implementation of an inference attack targeting reconstruction of the sensitive attributes of a model's training set, leveraging information regarding this model's fairness
 - Source code: <https://github.com/ferryjul/SensitiveAttributesReconstructionCorrector>

Academic Service

International Conferences

- **FAccT 2023, 2024, 2025, 2026** - ACM Conference on Fairness, Accountability and Transparency - *PC member*
- **ICML 2026** - International Conference on Machine Learning - *Reviewer*
- **AAAI 2022, 2026** - AAAI Conference on Artificial Intelligence - *PC member*
- **NeurIPS 2025, 2026** - Annual Conference on Neural Information Processing Systems - *Reviewer*
- **ECAI 2025** - European Conference on Artificial Intelligence - *PC member*
- **ECML PKDD 2024, 2025** - European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases - *PC member*
- **SaTML 2024** - 2nd IEEE Conference on Secure and Trustworthy Machine Learning - *PC member*
- **CP 2021, 2022, 2023** - International Conference on Principles and Practice of Constraint Programming - *Reviewer*
- **IJCAI 2022, 2023** - International Joint Conference on Artificial Intelligence - *Reviewer*

- **CPAIOR 2022, 2023, 2025** - International Conference on the Integration of Constraint Programming, Artificial Intelligence, and Operations Research - *Reviewer*

Journals

- **Neural Networks** - *reviewer*
- **Transactions on Knowledge and Data Engineering (TKDE)** - *reviewer*
- **Transactions on Machine Learning Research (TMLR)** - *reviewer*
- **Journal of Machine Learning Research (JMLR)** - *reviewer*

National Conferences

- **Canadian AI 2026** - 39th Canadian Conference on Artificial Intelligence - *reviewer*
- **APVP 2026** - 16^{ème} Atelier sur la Protection de la Vie Privée - *PC member*
- **APVP 2024** - 14^{ème} Atelier sur la Protection de la Vie Privée - *PC member*

Workshops

- **NeurIPS 2024 Workshop**: Workshop on Regulatable Machine Learning (**RegML**) - *reviewer*
- **NeurIPS 2024 Workshop**: Algorithmic Fairness Through the Lens of Metrics and Evaluation (**AFME**) - *reviewer*
- **NeurIPS 2023 Workshop**: Workshop on Regulatable Machine Learning (**RegML**) - *reviewer*
- **NeurIPS 2023 Workshop**: Algorithmic Fairness through the Lens of Time (**AFT**) - *reviewer*
- **AAAI 2023 Workshop**: The 4th AAAI Workshop on Privacy Preserving Artificial Intelligence (**PPAI**) - *reviewer*
- **NeurIPS 2022 Workshop**: Algorithmic Fairness through the Lens of Causality and Privacy (**AFCP**) - *reviewer*

Scientific Community

March 31–April 1, 2026	Jury Member at PolyCongRÉ, the Graduate Engineering Research Conference of Polytechnique Montréal
Since 2024	Member of the CIRRELT Reading Group
2021 - 2024	Enrolled in the NSERC (<i>Natural Sciences and Engineering Research Council of Canada</i>) CREATE (<i>Collaborative Research and Training Experience</i>) interdisciplinary training initiative on Responsible AI
April 2023	Member of the organizing committee for the LAAS-CNRS Decision & Optimization Department Workshop 2023
2022 - 2023	Member of the <i>LAAS PhD events</i> team - Organization of events for LAAS-CNRS PhD students (e.g., LAAS-CNRS doctoral students' day 2022)
2021 - 2023	PhD student representative (doctoral school MITT) - Member of the doctoral school's executive board and council
2021 - 2023	Member of the ACM Special Interest Group on Information Retrieval (SIGIR)
November 2020	Participation in the organization of a constraint programming hackathon for the International Autumn School on Constraint Programming, Combinatorial Optimization and Machine Learning organized by the ACP, ANITI, and the GDRs IA and RO of the CNRS

Supervision

Since October 2025	Co-supervising the PhD of Ziba Jabbar Zare at Polytechnique Montréal and Concordia University: <i>Fairness in Hybrid Interpretable Models</i> (with Thibaut Vidal)
Since September 2023	Co-supervising the internship and then the PhD of Awa Khouna at Polytechnique Montréal: <i>Efficient near-optimal counterfactual explanations and privacy attacks based on model explanations</i> (with Thibaut Vidal)
July 2025 - August 2025	Co-supervising the internship (Master 1) of Bastien Chaume at LAAS-CNRS : <i>Integrating privacy-preserving criteria within greedy tree induction algorithms</i> (with Marie-José Huguet)
March 2025 - August 2025	Co-supervising the internship (Master 2) of Homère Tandeau de Marsac at Polytechnique Montréal: <i>Efficient algorithms for heuristic near-optimal counterfactual explanations</i> (with Thibaut Vidal)
January 2025 - May 2025	Co-supervising the internship (Master 1) of Marie Leclerc at UQAM (Montréal, Canada) and IRISA (Rennes, France): <i>Leveraging tree-based local explanations of Graph Neural Networks to conduct dataset reconstruction attacks</i> (with Sébastien Gambs and Tristan Allard)
March 2024 - July 2024	Co-supervising the internship (Master 1) of Alice Gorgé at Polytechnique Montréal: <i>Investigating privacy-preserving mechanisms to prevent reconstruction attacks</i> (with Thibaut Vidal) For this internship, Alice received the Louis-Edouard Rivot Medal, awarded to students at École Polytechnique (X) who have distinguished themselves through the excellence of their research work.
February 2024 - June 2024	Co-supervising the internship (Master 2) of Lucas Langlade at Polytechnique Montréal: <i>Characterizing fairness and sparsity within the sets of good models</i> (with Thibaut Vidal and Gabriel Laberge)
August 2023 - February 2024	Co-supervising the internship (Master 2) of Timothée Ly at LAAS-CNRS: <i>Interpretable and privacy-preserving machine learning</i> (with Sébastien Gambs and Marie-José Huguet)
February - July 2023	Co-supervising the internship (Master 2) of Timothée Pascal at Polytechnique Montréal: <i>Dataset Reconstruction Attacks against tree-based models</i> (with Thibaut Vidal)
February - July 2022	Co-supervising the internship (Master 2) of Julien Rouzot at LAAS-CNRS: <i>Fairness in Multi-Class Classification</i> (with Marie-José Huguet)

Miscellaneous

Academic Visits

June - July 2022	École de Technologie Supérieure, Montréal, Canada
May - June 2022	Université du Québec à Montréal, Montréal, Canada

Research Projects

October 2021 - October 2023	CIMI PRIMAL project (<i>Operational Research for Fairness, Privacy and Interpretability in Machine Learning</i>), led by Marie-José Huguet (LAAS-CNRS) and launched on October 15, 2021. The project's members are: Julien Ferry, Mohamed Siala (LAAS-CNRS), Sébastien Gambs (Université du Québec à Montréal), and Ulrich Aïvodji (École de Technologie Supérieure, Montréal). 20,000€
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Others

2024-2025	External collaborator for the BMDec (Breizh Mobilité Décarbonée) scientific project, led by a consortium of four partners: the consultancy firm SETUR, RatpDev, and two research laboratories from the University of Rennes 1: CREM and IRISA. The project aims to design a decision-support tool for multimodal mobility that addresses the challenges of decarbonizing peri-urban "home-to-work" travel at the scale of the Brittany region. I have developed privacy audit tools to control the amount of information leaked by the released statistics and models regarding the collected user data.
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Publications

Journals

- [Akk+25] Fabian Akkerman, **Julien Ferry**, Christian Artigues, Emmanuel Hébrard, and Thibaut Vidal. "Boosting Revisited: Benchmarking and Advancing LP-Based Ensemble Methods". In: *Transactions on Machine Learning Research*. TMLR. 2025.
- [Fer+25] **Julien Ferry**, Ulrich Aïvodji, Sébastien Gambs, Marie-José Huguet, and Mohamed Siala. "Taming the Triangle: On the Interplays between Fairness, Interpretability and Privacy in Machine Learning". In: *Computational Intelligence*. Vol. 41. 4. Wiley, 2025, e70113.
- [FLA24] **Julien Ferry**, Gabriel Laberge, and Ulrich Aïvodji. "Learning Hybrid Interpretable Models: Theory, Taxonomy, and Methods". In: *Transactions on Machine Learning Research*. TMLR. 2024.
- [Fer+23c] **Julien Ferry**, Ulrich Aïvodji, Sébastien Gambs, Marie-José Huguet, and Mohamed Siala. "Improving Fairness Generalization Through a Sample-Robust Optimization Method". In: *Machine Learning, S.I. on Safe And Fair Machine Learning*. Vol. 112. 6. Springer Verlag, 2023, pp. 2131–2192.

International Conferences (with proceedings)¹

- [God+26] Augustin Godinot, Sofiane Azogagh, **Julien Ferry**, and Sébastien Gambs. "Manipulation-Proof Oblivious Audits against Deceptive Model Providers". In: *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society, AIES*. 2026 acceptance rate: 22.0% (220/990).
- [Gor+26] Alice Gorgé, **Julien Ferry**, Sébastien Gambs, and Thibaut Vidal. "Training Set Reconstruction from Differentially Private Forests: How Effective is DP?" In: *Proceedings of the 4th IEEE Conference on Secure and Trustworthy Machine Learning, SaTML*. 2026 acceptance rate: 26.2% (61/233).
- [Jab+26] Ziba Jabbar Zare, Ulrich Aïvodji, **Julien Ferry**, and Thibaut Vidal. "When Interpretability Is Unequally Distributed: Fairness in Hybrid Interpretable Models". In: *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society, AIES*. 2026 acceptance rate: 22.0% (220/990).
- [KfV26] Awa Khouna, **Julien Ferry**, and Thibaut Vidal. "Counterfactual Maps: What They Are and How to Find Them". In: *Proceedings of the 40th Annual Conference on Neural Information Processing Systems, NeurIPS*. 2026 acceptance rate: 25.7% (7900/30709).
- [KfV25] Awa Khouna, **Julien Ferry**, and Thibaut Vidal. "From Counterfactuals to Trees: Competitive Analysis of Model Extraction Attacks". In: *Proceedings of the 39th Annual Conference on Neural Information Processing Systems, NeurIPS*. 2025 acceptance rate: 24.5% (5290/21575), spotlight: 3.2% (688/21575).
- [Lan+25] Lucas Langlade, **Julien Ferry**, Gabriel Laberge, and Thibaut Vidal. "Fairness and Sparsity within Rashomon sets: Enumeration-Free Exploration and Characterization". In: *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society, AIES*. 2025 acceptance rate: 35% (239/683).

¹For [Gor+26], Alice Gorgé and Julien Ferry contributed equally and are co-first authors, as acknowledged in the paper.

For [Aiv+21], authors listed in alphabetical order (**first author bold**).

- [Fer+24a] **Julien Ferry**, Ulrich Aïvodji, Sébastien Gambs, Marie-José Huguet, and Mohamed Siala. “Probabilistic Dataset Reconstruction from Interpretable Models”. In: *Proceedings of the 2nd IEEE Conference on Secure and Trustworthy Machine Learning, SaTML*. 2024, pp. 1–17 acceptance rate: 21.5% (34/158).
- [Fer+24b] **Julien Ferry**, Ricardo Fukasawa, Timothée Pascal, and Thibaut Vidal. “Trained Random Forests Completely Reveal your Dataset”. In: *Proceedings of the 41st International Conference on Machine Learning, ICML*. 2024, pp. 13545–13569 acceptance rate: 27.5% (2610/9473), oral presentation: 1.5% (144/9473).
- [Fer+23b] **Julien Ferry**, Ulrich Aïvodji, Sébastien Gambs, Marie-José Huguet, and Mohamed Siala. “Exploiting Fairness to Enhance Sensitive Attributes Reconstruction”. In: *Proceedings of the 1st IEEE Conference on Secure and Trustworthy Machine Learning, SaTML*. 2023, pp. 18–41 acceptance rate: 26.3% (40/152).
- [Fer+23d] **Julien Ferry**, Ulrich Aïvodji, Sébastien Gambs, Marie-José Huguet, and Mohamed Siala. “Improving Fairness Generalization Through a Sample-Robust Optimization Method”. In: *The 37th AAAI Conference on Artificial Intelligence*. Poster and oral presentation of our 2022 Machine Learning Journal paper [Fer+23c]. AAAI 2023 (Journal Track). 2023.
- [Fer+22b] **Julien Ferry**, Ulrich Aïvodji, Sébastien Gambs, Marie-José Huguet, and Mohamed Siala. “Leveraging Integer Linear Programming to Learn Optimal Fair Rule Lists”. In: *Proceedings of the 19th International Conference on the Integration of Constraint Programming, Artificial Intelligence, and Operations Research, CPAIOR*. 2022, pp. 103–119 acceptance rate: 46.7% (28/60).
- [RFH22] Julien Rouzot, **Julien Ferry**, and Marie-José Huguet. “Learning Optimal Fair Scoring Systems for Multi-Class Classification”. In: *Proceedings of the 34th IEEE International Conference on Tools with Artificial Intelligence, ICTAI*. 2022, pp. 197–204 acceptance rate: 29% (217/747).
- [Aïv+21] Ulrich Aïvodji, **Julien Ferry**, Sébastien Gambs, Marie-José Huguet, and Mohamed Siala. “FairCORELS, an Open-Source Library for Learning Fair Rule Lists”. In: *Proceedings of the 30th ACM International Conference on Information & Knowledge Management, CIKM*. 2021, pp. 4665–4669 acceptance rate (Demo track): 43.2% (35/81).

Preprints²

- [Akk+26] Fabian Akkerman, **Julien Ferry**, Théo Guyard, and Thibaut Vidal. “Pace: Prune-And-Compress Ensemble Models”. In: *arXiv preprint arXiv:2605.06278 (under review)* (2026).
- [Kho+26] Awa Khouna, Youssouf Emine, **Julien Ferry**, and Thibaut Vidal. “Optimal Counterfactual Search in Tree Ensembles: A Study Across Modeling and Solution Paradigms”. In: *arXiv preprint arXiv:2605.06561 (under review)* (2026).
- [Tan+26] Homère Tandeau de Marsac, Awa Khouna, **Julien Ferry**, and Thibaut Vidal. “Walking the Forest: Region-Graph Search for Counterfactual Explanations”. In: *preprint upcoming (under review)* (2026).
- [VF26] Thibaut Vidal and **Julien Ferry**. “Trustworthy Machine Learning through the Lens of Combinatorial Optimization: Survey and Research Perspectives”. In: *arXiv preprint arXiv:2607.07762 (under review)* (2026).
- [Ly+24] Timothée Ly, **Julien Ferry**, Ulrich Aïvodji, Sébastien Gambs, and Marie-José Huguet. “Smooth Sensitivity for Learning Differentially-Private yet Accurate Rule Lists”. In: *arXiv preprint arXiv:2403.13848* (2024).
- [Aïv+19] Ulrich Aïvodji, **Julien Ferry**, Sébastien Gambs, Marie-José Huguet, and Mohamed Siala. “Learning fair rule lists”. In: *arXiv preprint arXiv:1909.03977* (2019).

National Conferences (with proceedings)

- [Fer+23a] **Julien Ferry**, Ulrich Aïvodji, Sébastien Gambs, Marie-José Huguet, and Mohamed Siala. “Exploiter l’équité d’un modèle d’apprentissage pour reconstruire les attributs sensibles de son ensemble d’entraînement”. In: *Rencontres des Jeunes Chercheurs en Intelligence Artificielle*. 2023.

²For [Aïv+19], authors listed in alphabetical order (**first author bold**)

For [Akk+26], Fabian Akkerman, Julien Ferry and Théo Guyard contributed equally.

[Fer+21a] **Julien Ferry**, Ulrich Aïvodji, Sébastien Gambs, Marie-José Huguet, and Mohamed Siala. “Améliorer la généralisation de l’équité en apprentissage grâce à l’Optimisation Distributionnellement Robuste”. In: *Rencontres des Jeunes Chercheurs en Intelligence Artificielle*. 2021.

National Conferences (without proceedings)

- [Fer+23e] **Julien Ferry**, Ulrich Aïvodji, Sébastien Gambs, Marie-José Huguet, and Mohamed Siala. “Utiliser l’équité d’un modèle d’apprentissage pour reconstruire les attributs sensibles de son ensemble d’entraînement”. In: *24ème conférence ROADEF de la Société Française de Recherche Opérationnelle et Aide à la Décision*. 2023.
- [Fer+22a] **Julien Ferry**, Ulrich Aïvodji, Sébastien Gambs, Marie-José Huguet, and Mohamed Siala. “Concilier l’équité statistique et la précision en apprentissage machine interprétable grâce à la PLNE”. In: *23ème conférence ROADEF de la Société Française de Recherche Opérationnelle et Aide à la Décision*. 2022.
- [Fer+21b] **Julien Ferry**, Ulrich Aïvodji, Sébastien Gambs, Marie-José Huguet, and Mohamed Siala. “Optimisation Distributionnellement Robuste pour améliorer la généralisation de l’équité en apprentissage”. In: *22ème conférence ROADEF de la Société Française de Recherche Opérationnelle et Aide à la Décision*. 2021.

PhD Manuscript

[Fer23] **Julien Ferry**. “Addressing Interpretability, Fairness and Privacy in Machine Learning Through Combinatorial Optimization Methods”. Theses. UPS Toulouse, Oct. 2023.

Talks

September 2026	<i>Training Data Leakage and Machine Learning Models</i> - Workshop at the Polytechnique Montréal booth, ALL IN 2026 (Montréal, Canada)
May 2026	<i>Algorithmic Biases, explainability and privacy risks</i> - OBVIA’s Early-Career Researchers’ Conference on Law and Digital Technologies (Montréal, Canada)
May 2026	<i>Boosting Revisited: Benchmarking and Advancing LP-Based Ensemble Methods</i> - Optimization Days (Montréal, Canada)
September 2025	<i>Fairness and Sparsity within Rashomon Sets: Enumeration-Free Exploration and Characterization</i> - PrivSec/TISL Seminar 2025 (UQAM & ETS, Montréal, Canada)
May 2025	<i>Trained Random Forests Completely Reveal your Dataset</i> - Seminar of the Data Academy of the National Bank of Canada (online)
May 2025	<i>Training Set Reconstruction from Differentially Private Forests: How Effective is DP?</i> - Optimization Days (Montréal, Canada)
March 2025	<i>Confidentiality and Privacy Facing Machine Learning</i> - Science Café for the 2024 Discoveries of the Year of the Quebec Association of Science Communicators (Québec City, Canada)
December 2024	<i>Reconstruction Attacks against Differentially Private Random Forests: Can DP Protect your Data ?</i> - LAAS-CNRS ROC Team invited seminar (Toulouse, France)
November 2024	<i>On the Privacy Risks of Machine Learning Models: Trained Random Forests Completely Reveal your Dataset</i> - Privacy Workshop (UQAM, Montréal, Canada)
September 2024	<i>Trained Random Forests Completely Reveal your Dataset</i> - Invited Seminar of the Confiance.ai research program (online)
September 2024	<i>Reconstruction Attacks against Differentially Private Random Forests: Can DP Protect your Data ?</i> - PrivSec/TISL Seminar 2024 (UQAM & ETS, Montréal, Canada)

July 2024	<i>Trained Random Forests Completely Reveal your Dataset</i> - ICML 2024 Oral presentation (Vienna, Austria) - recording (or long version)
June 2024	<i>Trained Random Forests Completely Reveal your Dataset</i> - PrivSec team Seminar (UQAM & ETS, Montréal, Canada)
May 2024	<i>Trained Random Forests Completely Reveal your Dataset</i> - Invited Talk at the MobiliT.AI forum on AI for critical systems applied to mobility (Montréal, Canada)
April 2024	<i>On the Interplays between Interpretability, Fairness & Privacy in Machine Learning: Trained Random Forests Completely Reveal your Dataset</i> - IRT/Université Toulouse 1 Capitole Seminar (Toulouse, France)
April 2024	<i>Privacy Attacks Against Machine Learning Models: Trained Random Forests Completely Reveal your Dataset</i> - LAAS-CNRS ROC Team invited seminar (Toulouse, France) - recording
April 2024	<i>Probabilistic Dataset Reconstruction from Interpretable Models</i> - SATML 2024 (Toronto, Canada) - recording
October 2023	<i>Addressing Interpretability, Fairness & Privacy in Machine Learning Through Combinatorial Optimization Methods</i> - PhD Defense (LAAS-CNRS, Toulouse, France) - recording
September 2023	<i>Exploiting Fairness to Enhance Sensitive Attributes Reconstruction</i> - PrivSec team Seminar 2023 (UQAM & ETS, Montréal, Canada (online))
July 2023	<i>Exploiting Fairness to Enhance Sensitive Attributes Reconstruction</i> - RJCIA 2023 (Strasbourg, France)
February 2023	<i>Exploiting Fairness to Enhance Sensitive Attributes Reconstruction</i> - ROADEF 2023 (Rennes, France)
February 2023	<i>Exploiting Fairness to Enhance Sensitive Attributes Reconstruction</i> - SATML 2023 (Raleigh, North Carolina, USA) - recording
July 2022	<i>Operational Research for Fairness, Privacy and Interpretability in Machine Learning: Leveraging ILP to Learn Optimal Fair Rule Lists</i> - DS4DM Coffee Talks (Polytechnique Montréal) - recording
June 2022	<i>Leveraging Integer Linear Programming to Learn Optimal Fair Rule Lists</i> - CPAIOR 2022 (Los Angeles, California, USA) - recording
May 2022	<i>Operational Research for Fairness, Privacy and Interpretability in Machine Learning: Leveraging ILP to Learn Optimal Fair Rule Lists</i> - Golnoosh Farnadi research group seminar (MILA, Montréal, Canada)
May 2022	<i>Operational Research for Fairness, Privacy and Interpretability in Machine Learning: Leveraging ILP to Learn Optimal Fair Rule Lists</i> - LATECE seminar (UQAM, Montréal, Canada)
May 2022	<i>Leveraging Integer Linear Programming to Learn Optimal Fair Rule Lists</i> - PrivSec team Seminar 2022 (UQAM, Montréal, Canada)
February 2022	<i>Leveraging MILP to Conciliate Statistical Fairness and Accuracy in Interpretable ML: Learning Optimal Fair Rule Lists</i> - ROADEF 2022 (Lyon, France)
November 2021	<i>FairCORELS, an Open-Source Library for Learning Fair Rule Lists</i> - CIKM 2021 (Gold Coast, Queensland, Australia (online)) - recording
October 2021	<i>Distributionally Robust Optimization to Improve Fairness Generalization in Machine Learning</i> - LAAS-CNRS Decision & Optimization Department Workshop (Mauvezin, France)
September 2021	<i>Addressing Interpretability, Fairness and Privacy in ML Through Combinatorial Optimization Methods</i> - GDR IA Autumn School (IA ²) - Poster Session (Paris, France)
July 2021	<i>Distributionally Robust Optimization to Improve Fairness Generalization in Machine Learning</i> - RJCIA 2021 (Bordeaux, France (online))

- April 2021 | *Distributionally Robust Optimization to Improve Fairness Generalization in Machine Learning* - ROADEF 2021 (Mulhouse, France (online))
- December 2020 | *Bias in Machine Learning* - LAAS-CNRS ROC Team seminar (Toulouse, France)

Personal Experiences

Community Involvement

- 2014 - 2023 | Vice-president of the Olemps Festival Committee (Anim'Olemps)
- 2019 - 2020 | Member of the GEIser department association (INSA Toulouse)
- 2016 - 2020 | Treasurer and co-founder of the Amicale des Jeunes Olempiens

Others

- 2021 - 2023 | Beta-tester for the *Decathlon Connect* sports app
- 2017 | First aid certification (PSE1)
- 2017 | Preparation of collective meals (Head cook)